

Water Balance/Budget And Its Need

The water balance in simple term represents the evaluation of sources of water in a watershed in a unit time and its corresponding discharges (it includes all losses for runoff)

Which can be represented in continuity equation

Change in mass storage in unit time=Mass inflow-Mass outflow

$$Q_{in} - Q_{out} = \Delta V / t \quad \text{or} \quad \Delta V = (P - Q - E) \Delta t \quad (\text{eq. 1})$$

Where : ΔV = change in water storage

P = Precipitation , Q = outflow as Runoff E = net evaporation , Δt = time interval for which the water balance is to be determined.

Water Balance for a Catchment

Or

$$P-R-G-E-T=\Delta S$$

eq.2

P= Precipitation, R= Surface runoff, G=Net Groundwater flow out of catchment, E=evaporation T= Transpiration, ΔS = Change in storage

ΔS = includes three component

$$\text{Change in storage } S = S_s + S_{sm} + S_g$$

S_s = surface water storage

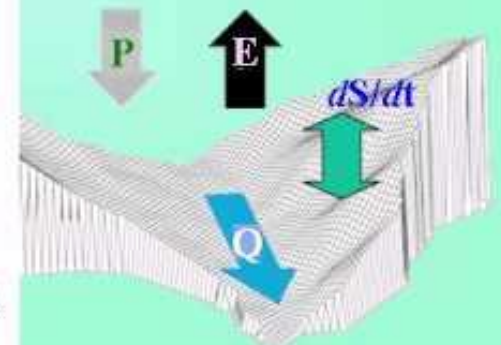
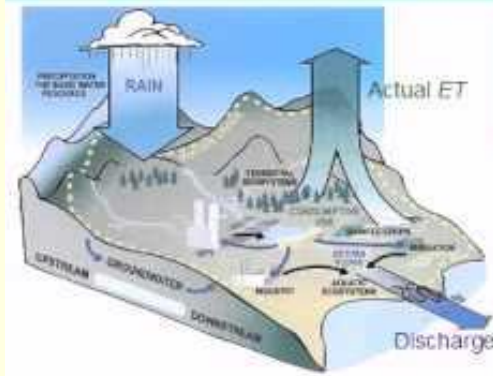
S_{sm} = Water storage as soil moisture

S_g = Water in storage as groundwater

Water Balance Equation

$$P = Q + E + dS/dt$$

P	:	Precipitation [mm a^{-1}]
Q	:	Discharge [mm a^{-1}]
E	:	Evaporation [mm a^{-1}]
dS/dt	:	Storage changes per time step [mm a^{-1}]



In terms of rainfall –runoff relationship Eq. 2 can be written as

$$R = P - L$$

L=losses (means water not available for surface runoff due to infiltration, evaporation, transpiration and surface storage)

In hydrological calculations, the volumes are often expressed as average depth over the catchment area.

What is catchment ?

Suppose Annual Runoff is 10^7 m^3 from the catchment area 10 km^2

Then corresponding depth , $D = \frac{10^7}{10 \times 10^6} = 1 \text{ m} = 100 \text{ cm}$

Exercise 1

A small watershed of 30 Km² received the rainfall of 15 cm in 90 minutes of rainfall event. Before the event of rainfall, there was no runoff at the outlet of the catchment. After the rainfall stream flow with an average discharge of 1.5 m³/sec for 10 hours. Again it dried up after the 10 hours. The answer followings

- I) Amount of losses which was not available to runoff
- II) What is ratio of runoff to precipitation

Exercise 2

WHY WATER BUDGET REQUIRED

Water Balance provide us contribution of different parameters

It is useful for the management and conservation of water resources

Water Resource Managers develop hydrological engineering models with economic development Models to track and predict supplies and demands

Important Component of Hydrological Cycle

Precipitation

Evaporation

Infiltration

Runoff

Precipitation its Measurement and Computation

Precipitation denotes all forms of water that reach the earth from the atmosphere.

The usual forms are rainfall, snowfall, hail, frost and dew. Rain dominates in all form.

Forms of Precipitation

- **1. Rain** - liquid deposits falling from the atmosphere to the surface
- with a diameter > 0.5 mm
 - < 0.5 mm: drizzle
 - max. size: about 5 - 7 mm
(too large to remain suspended)
 - beyond this size, inter-molecular cohesive forces become too weak to be held in the mass of water together as a single drop



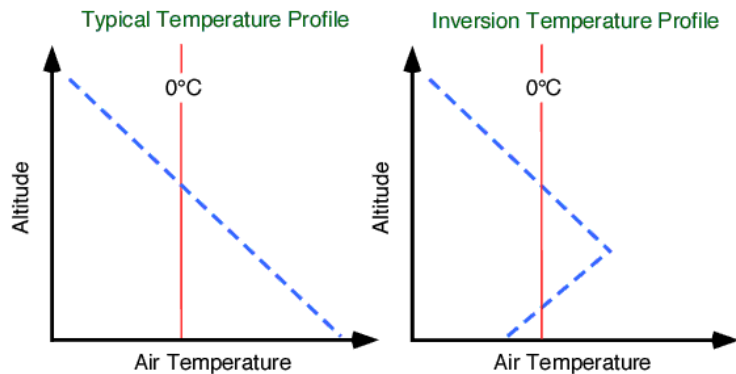
➤ **Raindrop size:**

drizzle	about 0.2 mm
light rain	about 0.5 mm
small rain	about 1.0 mm
rain	about 5.0 mm
rainstorm	up to about 7.0 mm

Forms of Precipitation

➤ 2. Freezing rain

- when falling liquid water droplets reaches a surface with a temperature below freezing point
- so, the rain droplets quickly turn into ice
- another * condition: where the rain develops, the temperature of rain develops must be above freezing



e.g. within a temperature inversion

Forms of Precipitation

■ 3. **Sleet / ice pellets**

- transparent / translucent spheres of frozen water
- with a diameter > 5 mm
- develop first as raindrops in relatively warm atmosphere (Temp: $>$ freezing),
- then raindrops descend into a colder layer of the atmosphere (Temp: $< 0^{\circ}\text{C}$)
- causing the freezing into ice pellets while reaching the ground surface
- like freezing rain, an air temperature inversion is required

Forms of Precipitation

➤ 4. Snow

- commonly found in the mid- and high- latitudes
- it develops when water vapour deposits itself directly to a six-sided (**hexagon**) deposition nuclei as a solid crystal, at temperature below freezing (what is this process called?)
- why is this unique form?
- snow is usually associated frontal uplifting with **mid-latitude cyclones**



Forms of Precipitation

➤ 5. Hail

- a frozen form of precipitation with a diameter > 5 mm
- hailstones: concentric shells of ice with alternating white cloudy appearance & those that are clear
- cloudy white: contain partially melted snowflakes that freeze on to the surface of the growing hailstone
- clear shell: develops when liquid water freezes onto the hailstone



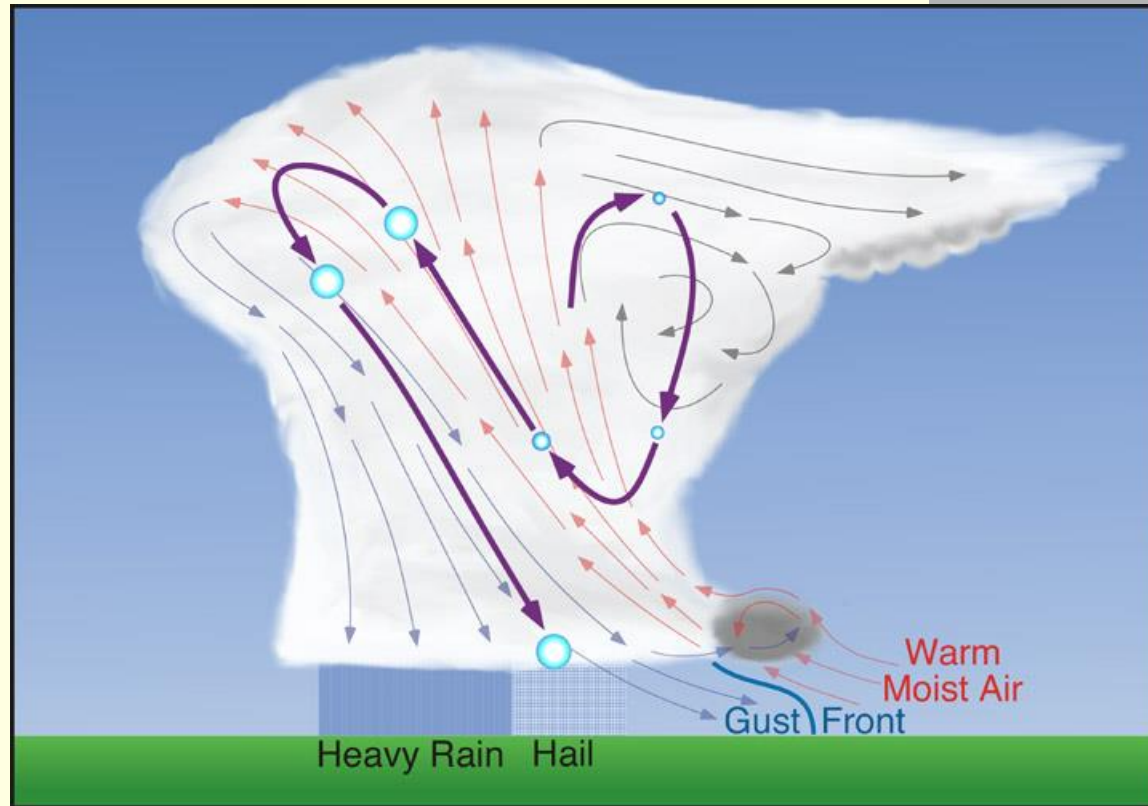
Forms of Precipitation

➤ 5. Hail

- Strong uprising currents in thunderstorm clouds provide the mechanism for forming hail
- the updrafts move hailstone embryos (e.g. large frozen raindrops) upward through the storm cloud
- where they encounter layers of ice crystals, snow & supercooled rain
- each encounter causes the hailstone to grow larger
- hailstones can grow very large in size with repeated updrafts
- until being too heavy to be supported by updrafts, it begins falling under gravity
- on descending, it can lose great mass due to melting

Forms of Precipitation

➤ 5. Hail



A typical hailstone growth path through a thunderstorm cloud. Hailstones add most of their mass during updrafts.

Necessary conditions?

- When clouds form in the atmosphere, they are non-precipitating in 99% of cases.



Necessary conditions?

- 1. cooling of air (e.g. convectional / orographic /cyclonic (frontal) uplifting)
 - 2. condensation and cloud formation
 - 3. an accumulation of moisture over nuclei (i.e. called CCN cloud condensation Nuclei)
 - 4. the growth of cloud droplets so that condensation can reach the earth
 - the most critical stage!!
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- **Water droplets and ice crystals** of clouds need to be transformed into heavier particles if they are to fall out of clouds as precipitation. **Why?**

Necessary conditions?

- **Water droplets and ice crystals** of clouds need to be transformed into heavier particles if they are to fall out of clouds as precipitation. **Why?**
- As we know, water droplets & ice crystals can only fall to the Earth's surface if they grow to a sufficiently large size that can overcome updrafts

Types of Precipitation:

(i) Thermal convection (convictional precipitation):

The air close to the warm earth gets heated and rises due to its low density, cools adiabatically to form a cauliflower shaped cloud, which finally bursts into a thunder storm. When accompanied by destructive winds, they are called 'tornados'.

(ii) Frontal Precipitation (Conflict between two air masses)

When two air masses due to contrasting temperatures and densities clash with each other, condensation and precipitation occur at the surface of contact. This surface of contact is called a 'front' or 'frontal surface'.

(iii) Orographic lifting (orographic precipitation)

The mechanical lifting of moist air over mountain barriers, causes heavy precipitation on the windward side. For example Cherrapunji in the Himalayan range and Agumbe in the western Ghats of south India get very heavy orographic precipitation of 1250 cm and 900 cm (average annual rainfall), respectively.

The wettest place is Mawsynram, Meghalaya, India, with 11,873 mm (467 in) of rain per annum

(iv) Cyclonic (cyclonic precipitation)

This type of precipitation is due to lifting of moist air converging into a low pressure belt, *i.e.*, due to pressure differences created by the unequal heating of the earth's surface. Here the winds blow spirally inward counterclockwise in the northern hemisphere and clockwise in the southern hemisphere. There are two main types of cyclones—tropical cyclone (also called hurricane or typhoon) of comparatively small diameter of 300-1500 km causing high wind velocity and heavy precipitation, and the extra-tropical cyclone of large diameter up to 3000 km causing wide spread frontal type precipitation.

Characteristics of Precipitation in India

South-west monsoon -----Period June to September (also called Monsoon)

75 to 80% of the total annual rainfall occur in major portion of the country
Originates from Indian Ocean

Post Monsoon --Period October --November

South west monsoon retreats, due to low pressure areas in Bay of Bengal and north easterly flow of the air that picks up moisture in Bay of Bengal. This air mass strikes east coast of Peninsula and causes rainfall.

Winter Precipitation --Period December to February also called western disturbances

Disturbances of extra tropical origin travel eastwards across Afghanistan and Pakistan (cause moderate to heavy rain and snowfall in Himalayas and northern plains)

Summer or Pre-monsoon Rain Period --March to May

Very little rainfall in India in this season (convective cells cause few thunderstorms mainly in Kerala, West Bengal and Assam)

Measurement

Precipitation is expressed in terms of the depth to which rainfall water would stand on an area if all the rain were collected on it. Thus 1 cm of rainfall over a catchment area of 1 km² represents a volume of water equal to 10⁴ m³.

Rainfall volume = watershed area (m²)* rainfall (m)

Annual Rainfall -----

Monthly Rainfall

Seasonal Rainfall

Average Annual rainfall of the country is 118.3 cm it varies from 10 cm in the western desert to 1100 cm in the north east region

Graphical Presentation of Rainfall Data

- i) **Mass Curve of rainfall** ---- Plot of accumulated precipitation vs time
- ii) **Hyetograph** -----plot of rainfall intensity vs time interval
Usually represented in bar chart
Represent the characteristics of a rain storm
- iii) **Point Rainfall**----- represent station rainfall

Instruments and Observational Methods of Rainfall

Non Recording Gauges

Standard Daily Raingauge

Recording Gauges

Tipping Bucket Type

Weighing-Bucket Type

Natural-Syphon Type

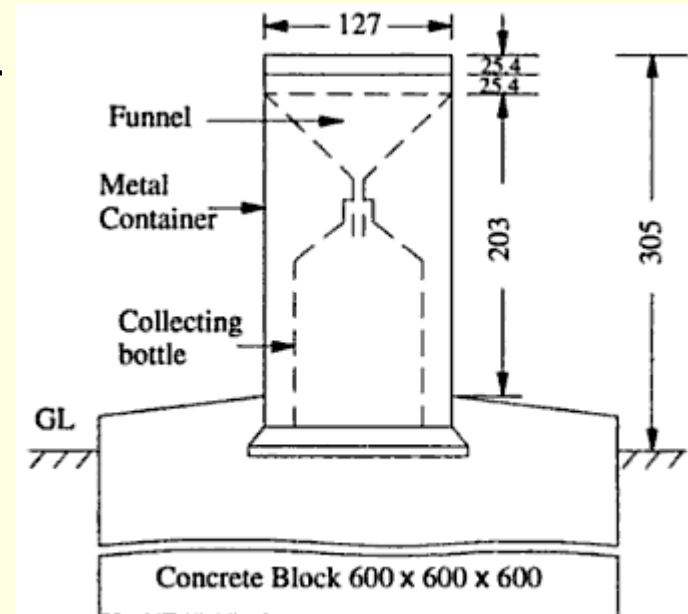
Telemetering Raingauges

Radar Measurement of Rainfall

- **Standard daily rain gauge**
- **Syphon gauge with chart recorder**
- **Tipping bucket gauge with digital recorder**

Standard Raingauge

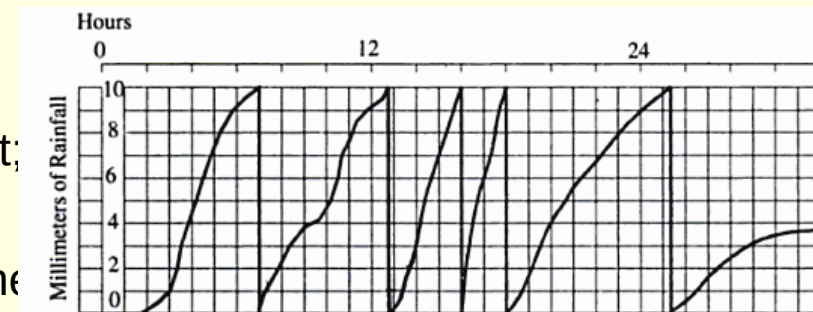
- A circular collector funnel with a brass or gun metal rim and a collection area of either 200 cm^2 or 100 cm^2 , leading to a, base unit, partly embedded in the ground and containing, a polythene collector bottle
- The gauge is read once or twice daily and any rain held in the polythene collector is poured into a measuring glass to determine rainfall in millimetres



Autographic Raingauge (Natural Syphon)

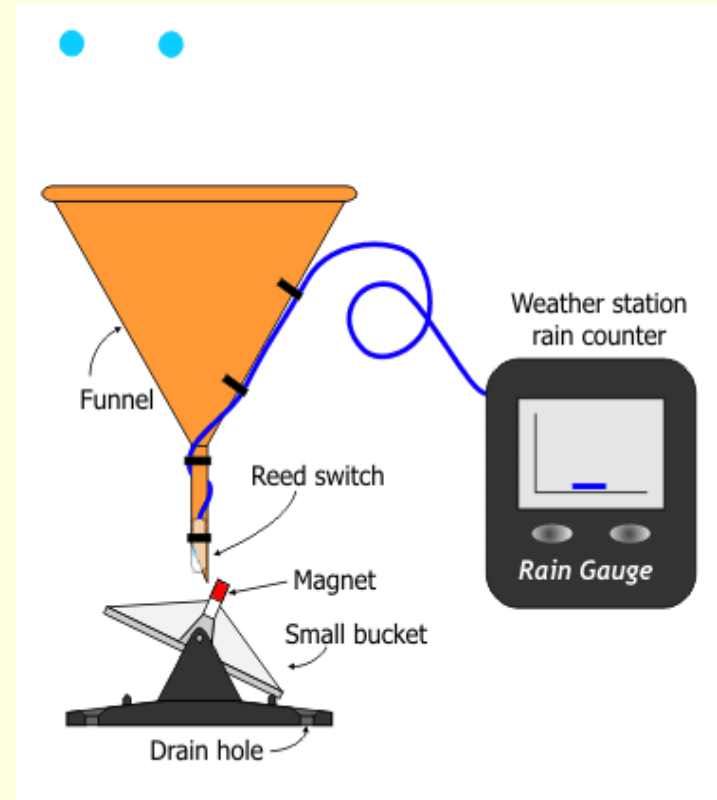
The natural syphon raingauge consists of a circular collector funnel with a gun metal rim of 200 mm diameter and set at 750 mm above ground level, leading to a float chamber. The float which rises with rainfall entering the chamber, a syphon chamber is attached to the float chamber and syphon action is initiated when the float rises to a given level. The float travel from syphon action to the next represents 10 mm rainfall, a float spindle projects from the top of the float to which is attached, a pen which records on a chart placed on a clock drum with a mechanical clock.

The chart is changed daily at the principal recording hour. During periods of dry weather the rainfall traces a horizontal line on the chart; during rainfall it produces a sloping line, the steepness of which defines the intensity of rainfall. The chart is graduated in hours and the observer extracts the hourly totals from the chart and enters it in a register and computes the daily total.



Tipping Bucket Raingauge

- Consists of a circular collector funnel with a brass or gunmetal rim of differing diameters leading to a tipping bucket arrangement which sits on a knife edge. It fills on one side then tips filling the second side and so on. A reed switch actuated by a magnet registers the occurrence of each tip. A logger records the occurrence of each tip and places a time stamp with the occurrence
- **Regular calibration is necessary**



Measurement of Snow

Depth of Snowfall

Measure and record the greatest amount of snowfall that has accumulated on your snowboard (wooden deck or ground if board is not available) since the previous snowfall observation. This measurement should be taken minimally once-a-day but can be taken up to four times a day, (every 6 hours) and should reflect the greatest accumulation of new snow observed (in inches and tenths, for example, 3.9 inches) since the last snowfall observation.

Water Equivalent of Snow

Water equivalent of snow is the depth of water that would result in melting of a unit of snow.

Snow Gauges

Like rain gauges, snow gauges are receptacles to catch precipitation as it fall in specified sampling area.